(Chapter 1)   19.1.26

Basic Concepts

1. **Current**: Electric current is the rate of flow of electric charge per unit time.

Formula —

$$I = \frac{\Delta q}{\Delta t} \quad (A / Cs^{-1})$$

$$\Rightarrow I = \frac{dq}{dt}$$

$$\Rightarrow dq = I\,dt$$

$$\Rightarrow q = \int_{t_1}^{t_2} I\,dt \quad (C)$$

where,
I = Current (Ampere)
q = total charge (Coulomb)
t = time (second)
Δq = change in charge
Δt = time interval (সময়ের পার্থক্য) $= (t_2 - t_1)$

2. **Voltage**: Voltage is the work done per unit charge to move a charge between two points.

Formula —

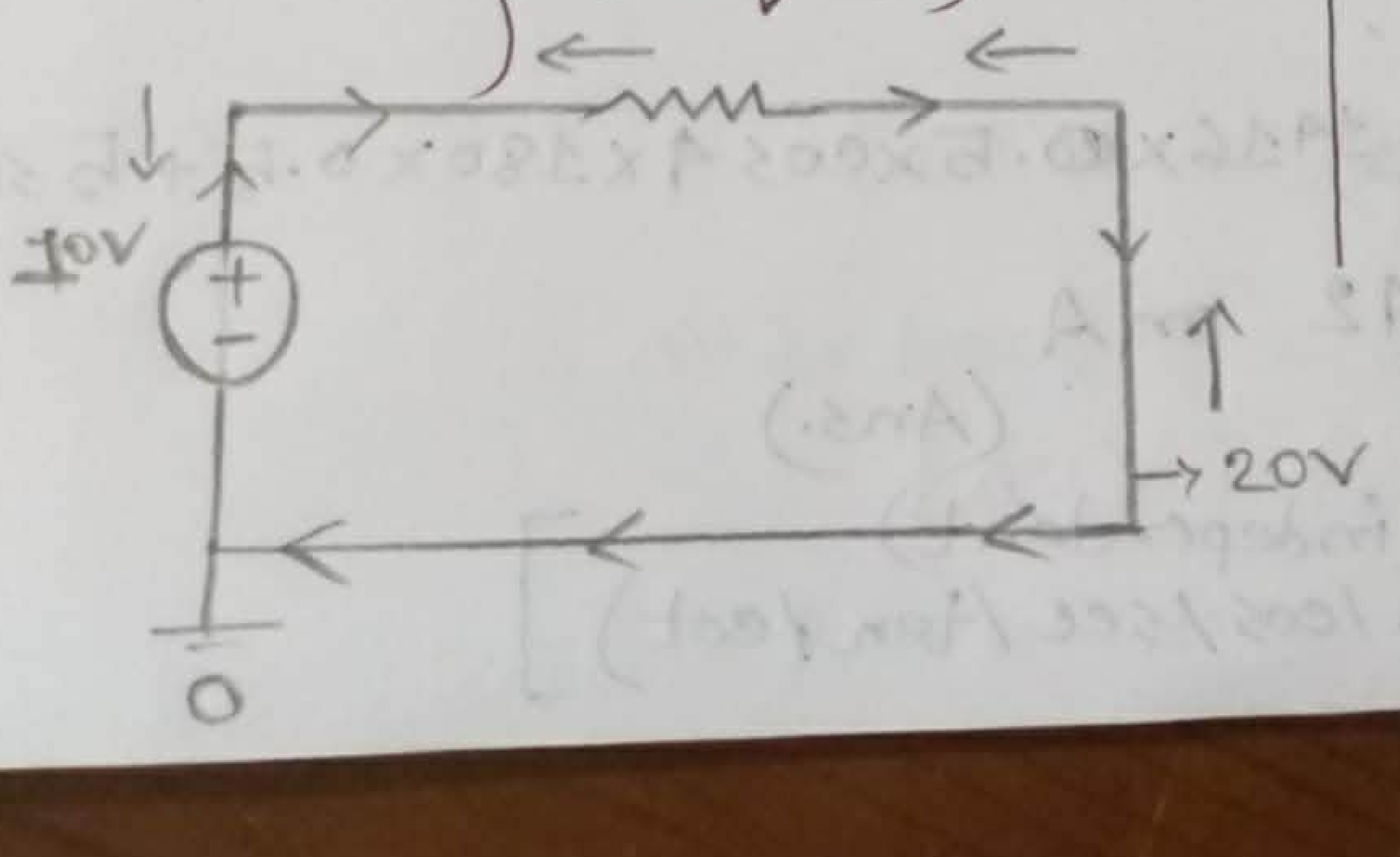
$$V = \frac{\Delta w}{\Delta q} \quad (V / JC^{-1})$$

$$\Rightarrow V = \frac{dw}{dq}$$

$$\Rightarrow dw = V\,dq$$

$$\Rightarrow w = \int V\,dq \quad (J)$$

where,
V = Voltage (volt)
w = work/Energy (Joule)
q = Charge (Coulomb)



[Higher to lower]

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3. Power : Electric power is the rate at which electrical energy is consumed or work is done.

Formula —

$$P = \frac{\Delta w}{\Delta t} \quad (W/Js^{-1})$$

$$\Rightarrow P = \frac{\Delta q}{\Delta t} \cdot \frac{\Delta w}{\Delta q}$$

$$= I \cdot V \quad (VA)$$

where,
P = Power (watt)
w = Energy (Joule)
t = time (sec)

Example —

1.2 Given that,

$$q = 5t \sin 4\pi t \ mC$$

We know,

$$I = \frac{\Delta q}{\Delta t}$$

$$= \frac{d}{dt}(5t \sin 4\pi t)$$

$$= 5t \cdot \cos 4\pi t \cdot 4\pi + \sin 4\pi t \cdot 5$$

$$= 20\pi t \cdot \cos 4\pi t + 5 \sin 4\pi t$$

At $t = 0.5s$,

$$I = 20 \times 3.1416 \times 0.5 \times \cos 4 \times 180 \times 0.5 + 5 \sin 4 \times 180 \times 0.5$$

$$= 31.42 \ mA$$

(Ans.)

[$\pi = 3.1416$ (independent)
$\pi = 180$ (sin/cos/sec/tan/cot)]

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1.3 Given that,

$t_1 = 1s$, $t_2 = 2s$, $I = (3t^2 - t)\ A$

We know,

$$q = \int_{t_1}^{t_2} I\, dt$$

$$= \int_1^2 (3t^2 - t)\, dt$$

$$= \left[3 \cdot \frac{t^3}{3} - \frac{t^2}{2}\right]_1^2$$

$$= \left(2^3 - \frac{2^2}{2}\right) - \left(1^3 - \frac{1^2}{2}\right)$$

$$= 8 - 2 - 1 + \frac{1}{2}$$

$$= 5.5\ C$$

(Ans.)

1.4 Given that,

$\Delta w = 2.3 \times 10^3\ J$, $I = 2A$, $\Delta t = 10s$

We know,

Voltage drop, $V = \frac{\Delta w}{\Delta q}$

$$= \frac{2.3 \times 10^3}{20}$$

$$= 115\ V$$

$\Delta q = I \cdot \Delta t$

$= 2 \times 10$

$= 20\ C$

(Ans.)

1.5 Given that, (a)

$I = 5\cos 60\pi t$ A

$t = 3ms$

$= 3\times10^{-3}s$

(a) $v = 3I$

(b) $v = 3\cdot\frac{dI}{dt}$

(a) We know,

$P = VI$

$= 3I\cdot I \quad [\because v = 3I]$

$= 3I^2$

$= 3\times(5\cos 60\pi t)^2$

$= 3\times 25\cos^2 60\pi t$

$= 75\cos^2 60\pi t$

At $t = 3\times10^{-3}s$

$P = 75\times\cos^2(60\times180\times3\times10^{-3})$

$= 53.467$ W [absorb Power]

(Ans.)

(b) $v = 3\frac{dI}{dt}$

$= 3\times\frac{d}{dt}(5\cos 60\pi t)$

$= 3\times 5\cdot -\sin 60\pi t\times 60\pi$

$= -900\pi\sin 60\pi t$

$\therefore P = VI = -900\pi\sin 60\pi t\times 5\cos 60\pi t$

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$= -1500 \times 3.1416 \times \sin(60 \times 180 \times 3 \times 10^{-3}) \times \cos(60 \times 180 \times 3 \times 10^{-3})$

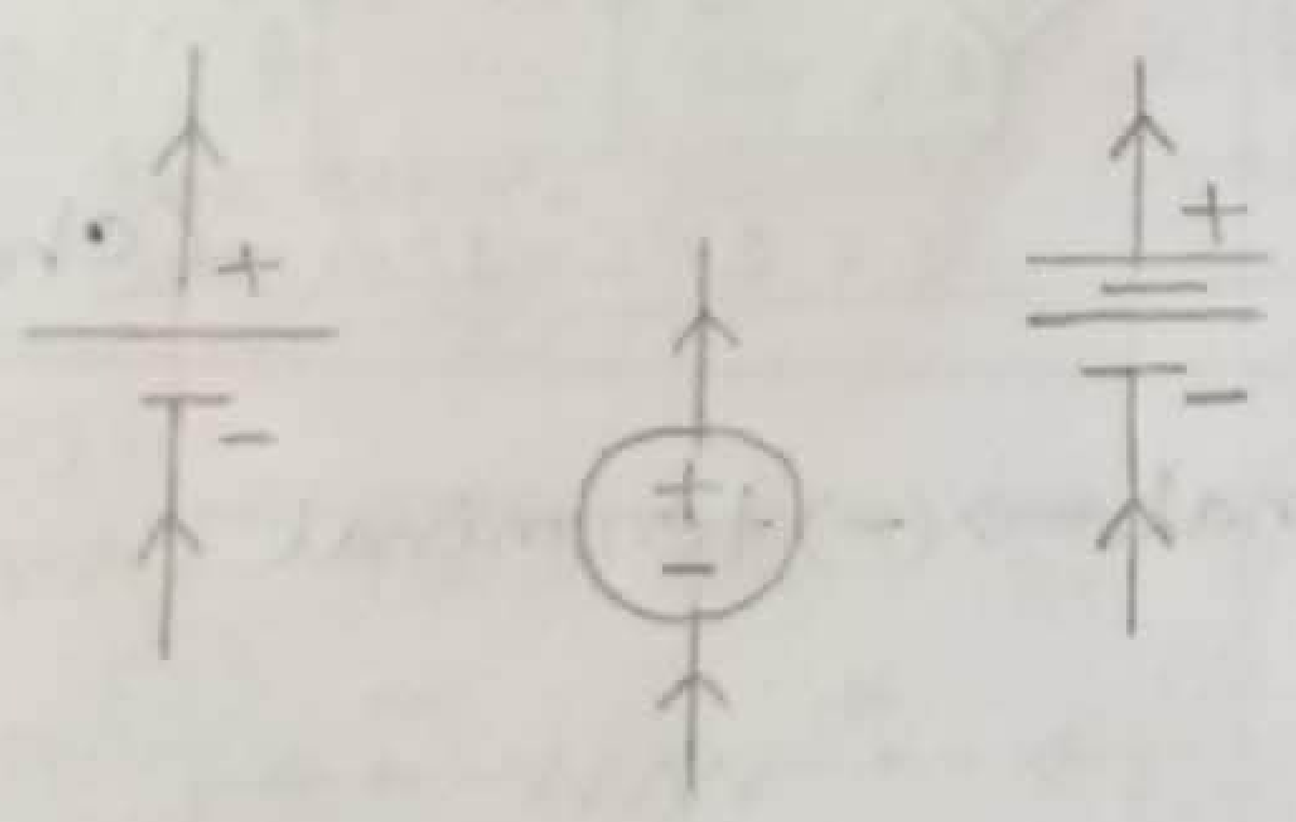
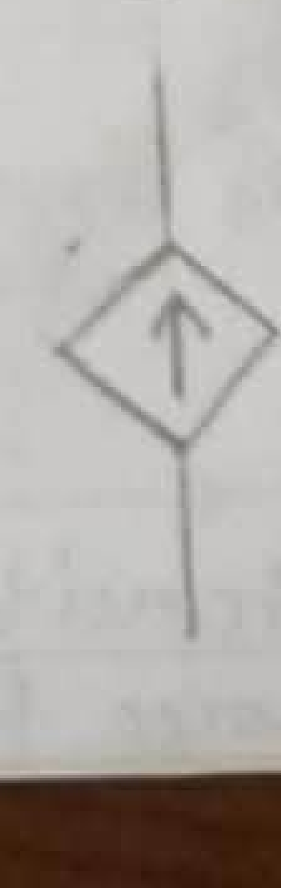
$[\because t = 3 \times 10^{-3} s]$

$= -6395.9$ W [Supply Power]

(Ans.)

$\boxed{\begin{array}{l} * \int 5\cos 60\pi t = \dfrac{5\sin 60\pi t}{60\pi} \\ * \dfrac{d}{dt}(5\cos 60\pi t) = -5\sin 60\pi t \cdot 60\pi \end{array}}$

- <u>DC Source</u>: It is a type of electric current that flows in only one direction and usually has constant voltage. (Direct current source)

(i) Voltage Source	(ii) Current Source
Independent (মানের পরিবর্তন হবেনা) 	Independent (মানের পরিবর্তন হবে না) 
Dependent (মানের পরিবর্তন হবে) 	Dependent (মানের পরিবর্তন হবে) 

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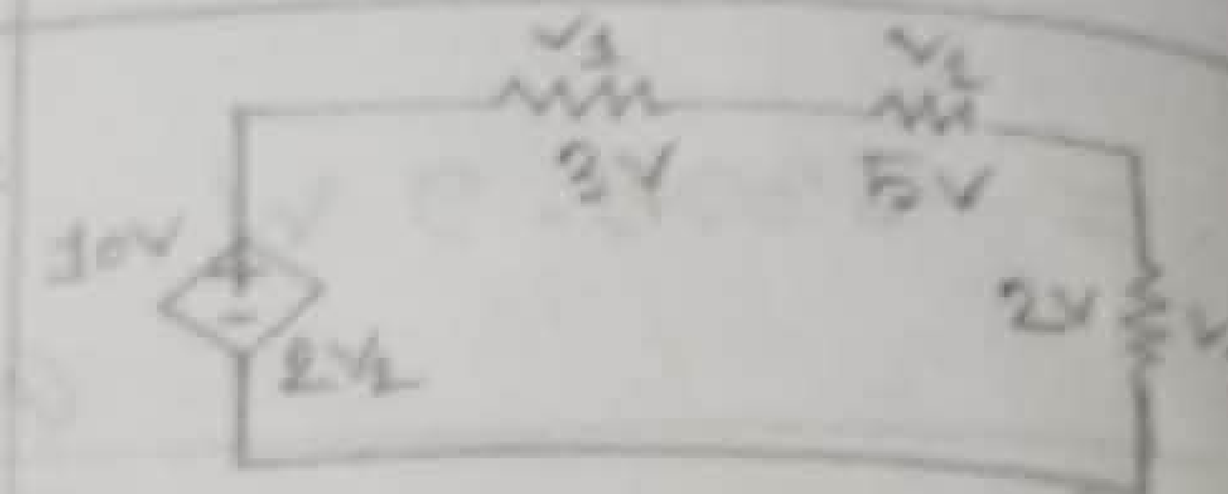
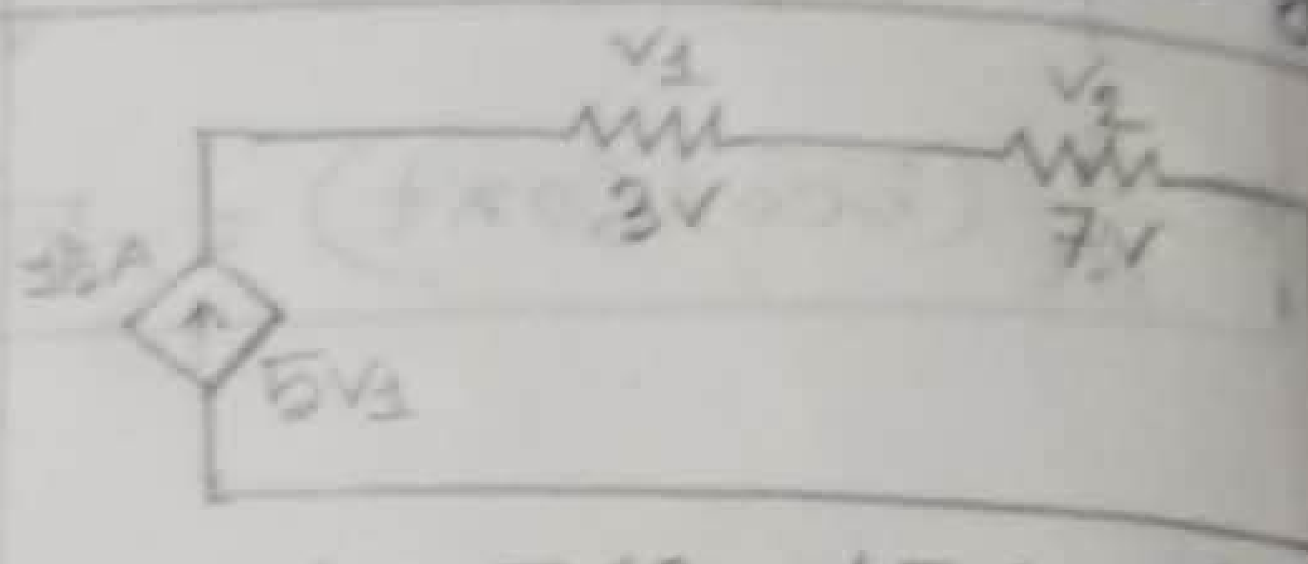
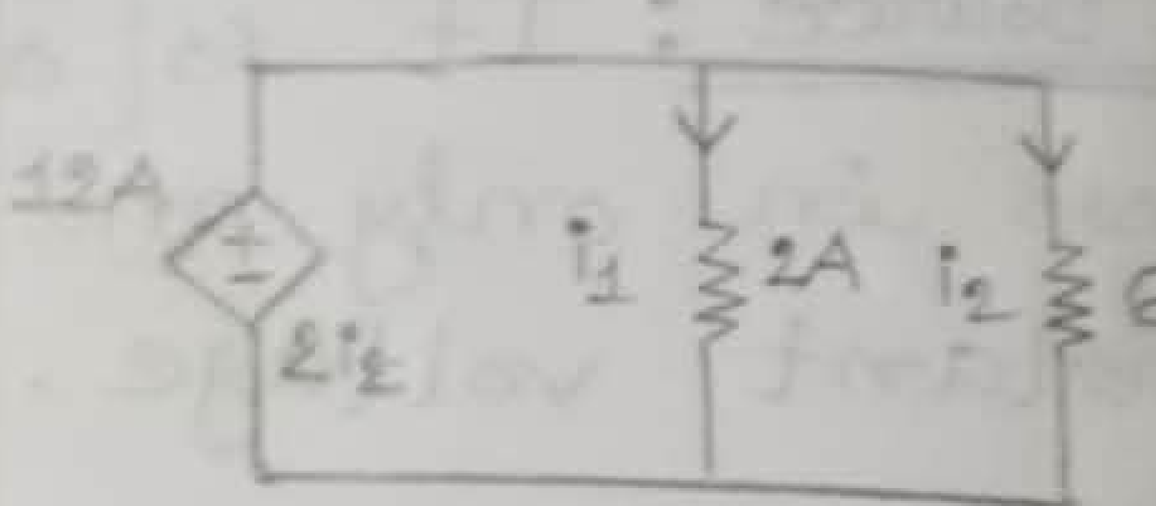
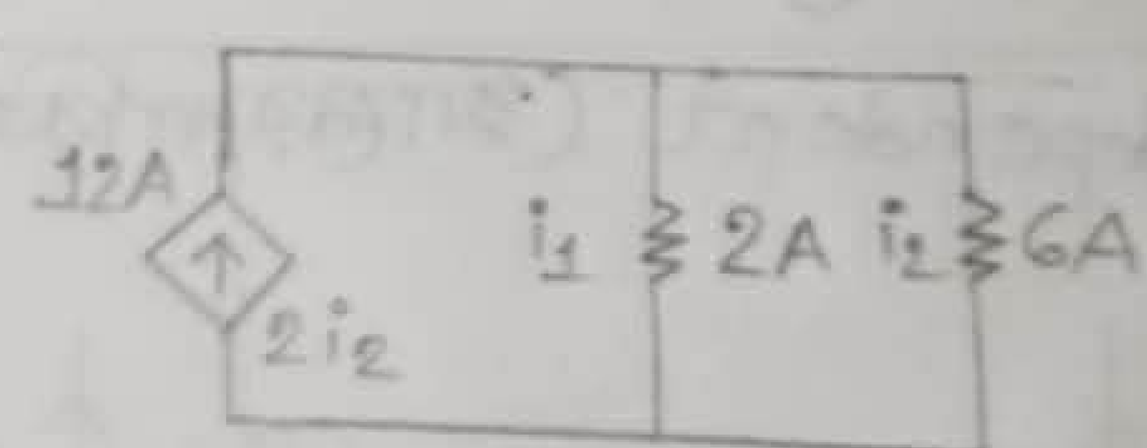
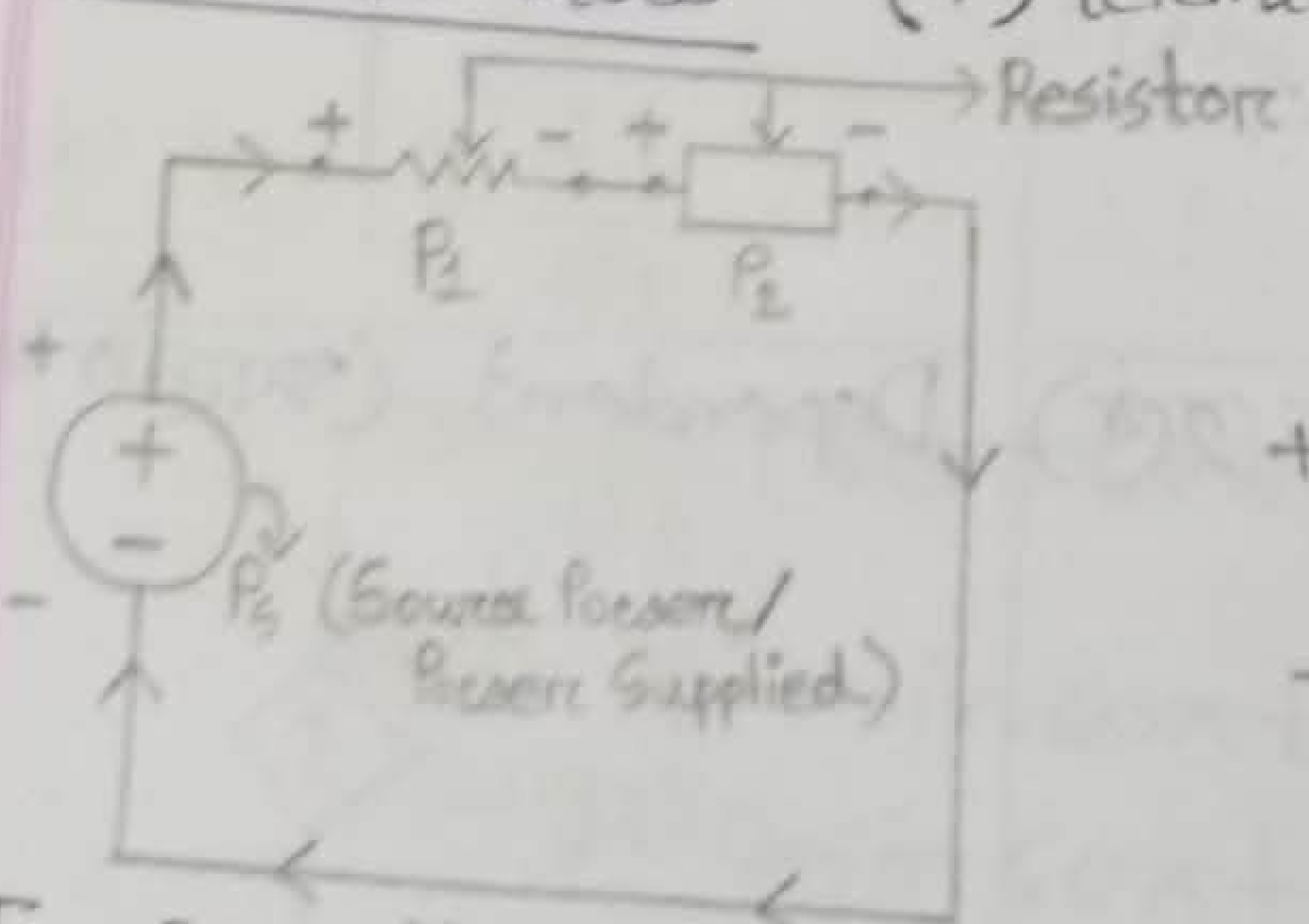
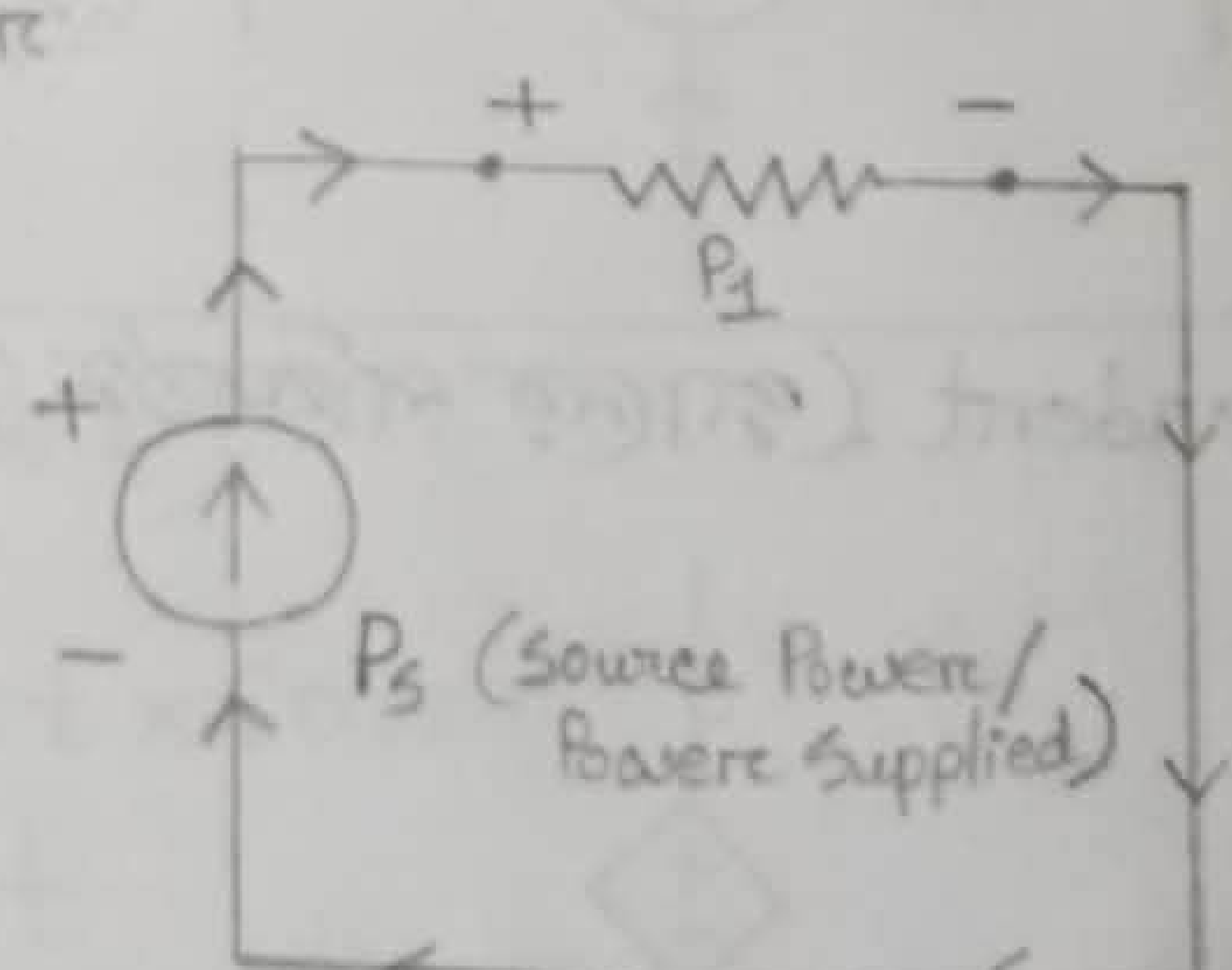
Name	Diagram
1. Voltage Controlled Voltage Source	 $V_2 = 2 \times 5 = 10v$ (Charge)
2. Voltage Controlled Current Source	 $V_1 = 5 \times 3 = 15A$ (Charge)
3. Current Controlled Voltage Source	 $i_2 = 2 \times 6 = 12A$ (Charge)
4. Current Controlled Current Source	 $i_2 = 2 \times 6 = 12A$ (Charge)

* Current flow - (+) terminal → (−) terminal





In Circuit,

$\boxed{\text{Source Power } (P_s) - \text{Absorbed Power } (P_1 + P_2 / P_1) = 0}$

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or,

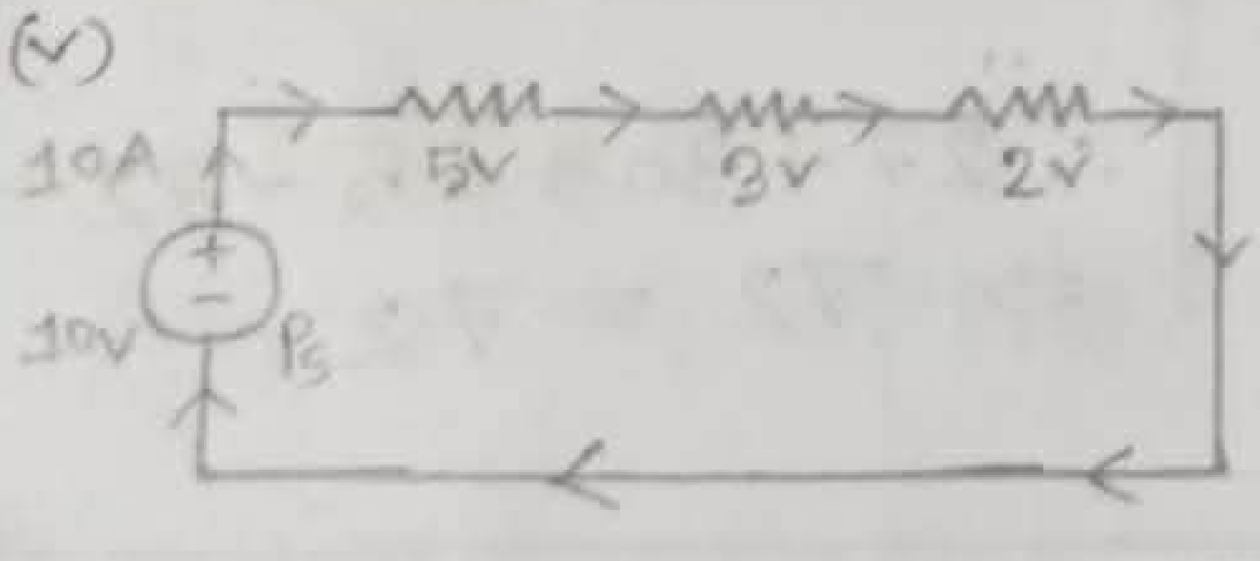
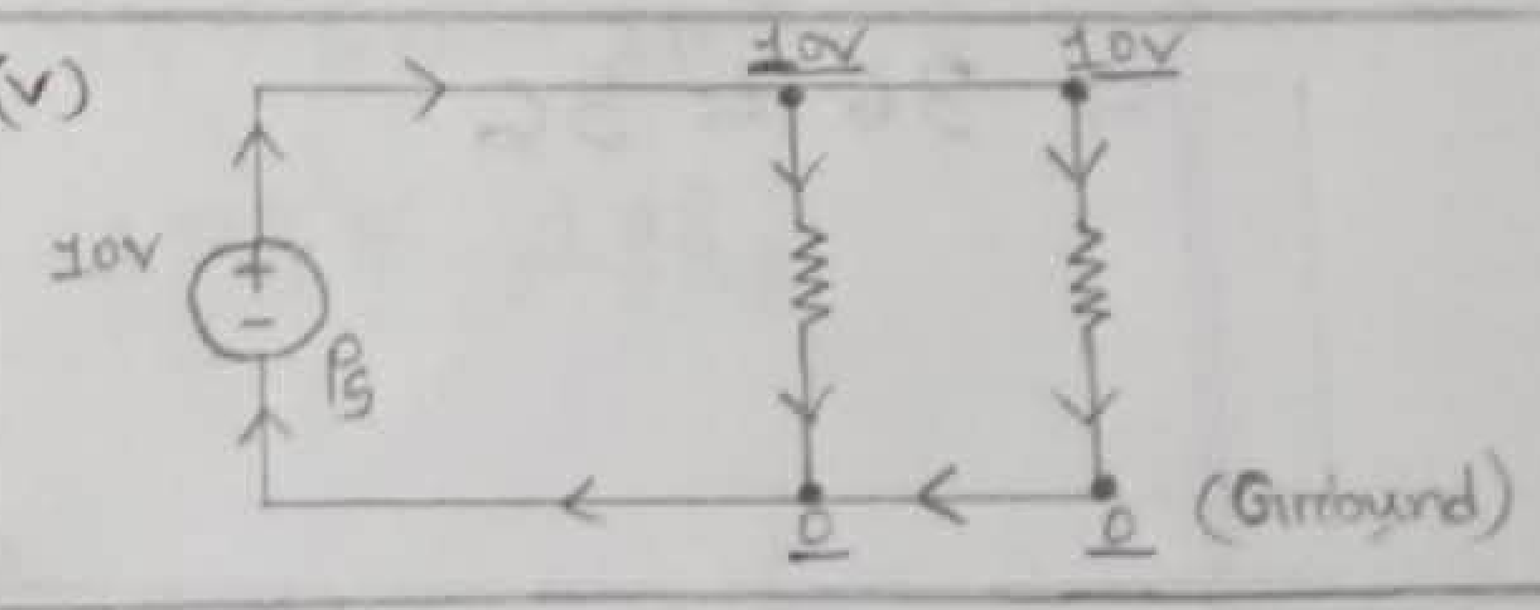
$\boxed{\text{Source Power} = \text{Absorbed Power}}$

- **Passive Sign Convention:**

(i) Current $(I) \rightarrow (-) \rightarrow (+) \rightarrow P = -VI \rightarrow$ Supply Power

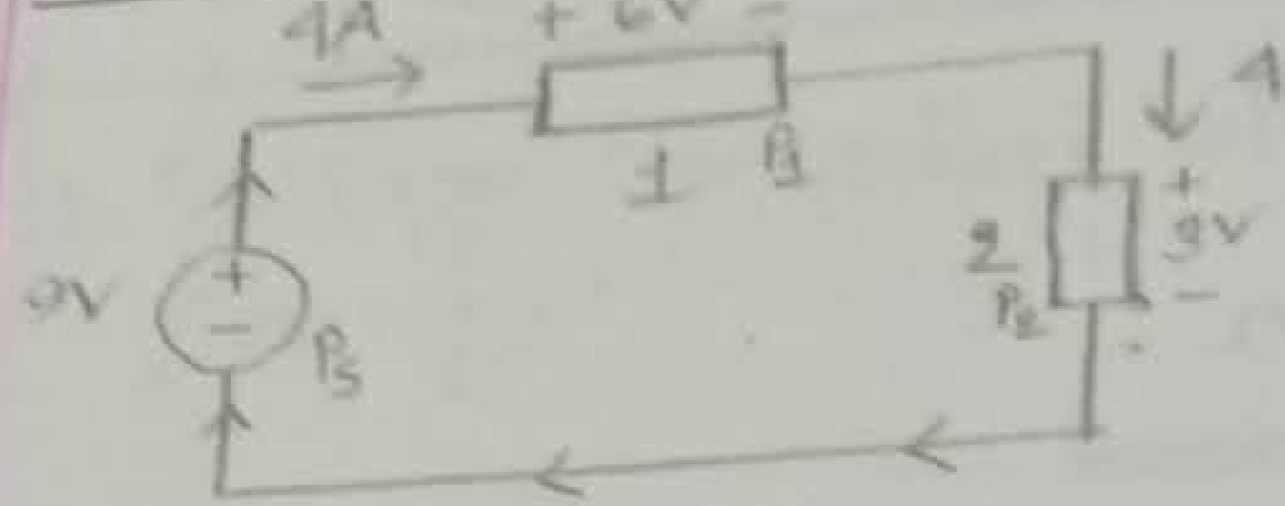
(ii) Current $(I) \rightarrow (+) \rightarrow (-) \rightarrow P = +VI \rightarrow$ Absorb Power

-

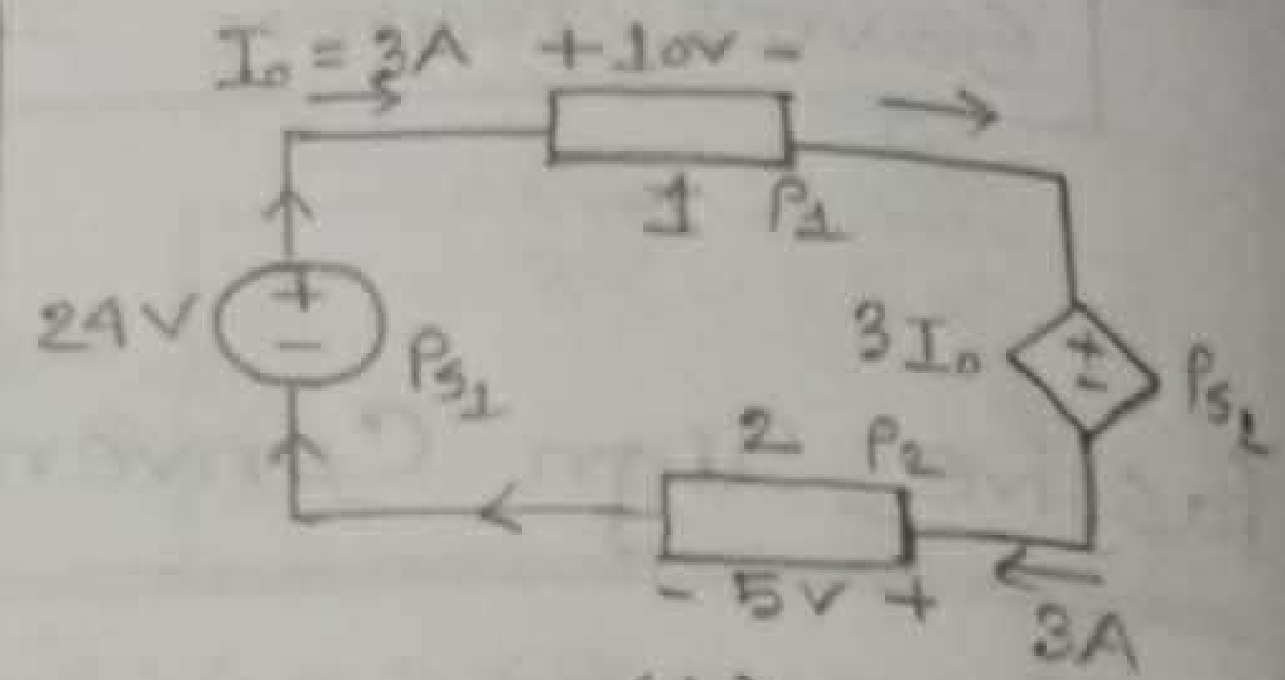
Series Connection -	Parallel Connection -
(i) Current Same $I_1 = I_2 = I_3$	(i) Current divided $I = I_1 + I_2 + I_3$
(ii) Voltage Divided $V = V_1 + V_2 + V_3$	(ii) Voltage Same (দুই প্রান্তের voltage সমান হবে).
(iii) $R_T = R_1 + R_2 + R_3$ ↳ Total Resistance	(iii) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
(iv) If one resistance fails, the whole circuit stop working.	(iv) If one resistance fails, other branches still work.
(v) 	(v) 

Example –

1.18)



(a)



(b)

$P_s = -VI$

$= -9\times4$

$= -36$ W (supply)

$P_1 = +VI$

$= 6\times4$

$= +24$ W (absorb)

$P_2 = +VI$

$= 3\times4$

$= +12$ W (absorb)

$\therefore 24+12-36=0$

$\Rightarrow 36 = 36$

$P_{s_1} = -VI$

$= -24\times3$

$= -72$ W (supply)

$P_{s_2} = +VI$

$= +3I_o\times3$

$= 3\times3\times3$

$= +27$ W (absorb)

$P_1 = +VI$

$= 10\times3$

$= +30$ W (absorb)

$P_2 = +VI$

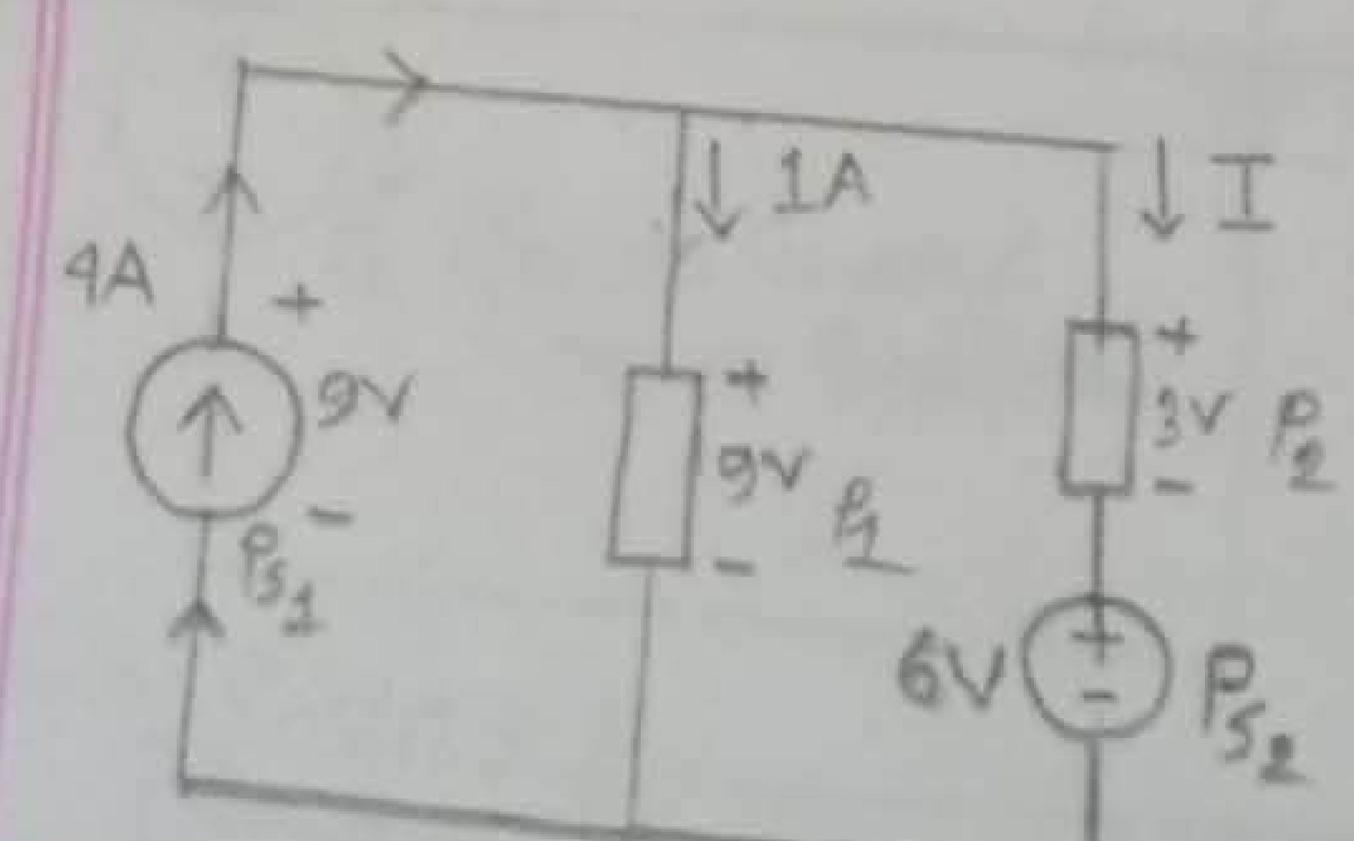
$= +5\times3$

$= +15$ W (absorb)

$\therefore 27+30+15-72=0$

$\Rightarrow 72 = 72$

1.19)



$P_{s_1} = -VI$

$= -9\times4$

$= -36$ W (supply)

$P_{s_2} = +VI$

$= +6\times I$

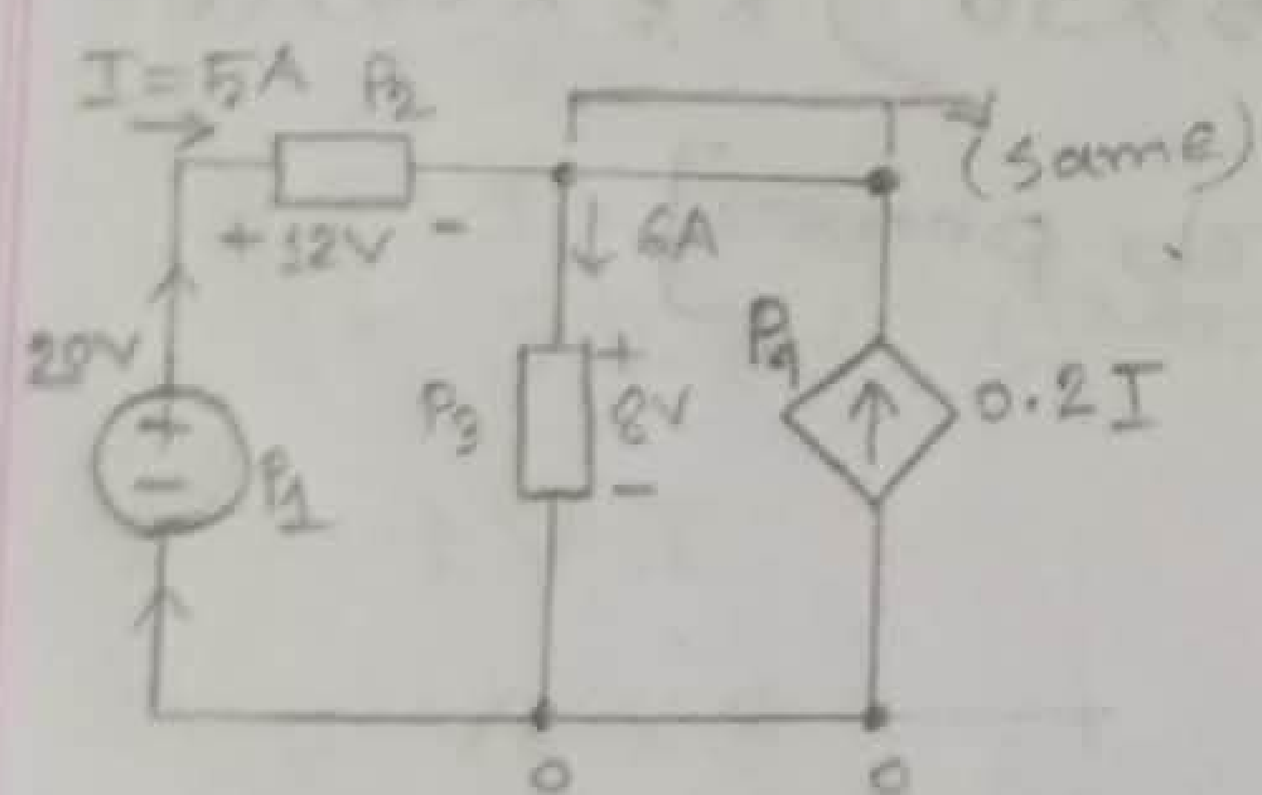
$= 6I$ (absorcb)

$P_1 = +VI$
$= +9 \times 1$
$= 9$ W (absorcb)

$P_2 = +VI$
$= +3 \times I$
$= 3I$ (absorcb)

$\therefore\ 3I + 9 + 6I - 36 = 0$
$\Rightarrow 9I - 27 = 0$
$\Rightarrow 9I = 27$
$\Rightarrow I = 3$ A

1.7



$P_1 = -VI$
$= -20 \times 5$
$= -100$ W (supply)

$P_2 = +VI$
$= 12 \times 5$
$= 60$ W (absorcb)

$P_3 = +VI$
$= 8 \times 6$
$= 48$ W (absorcb)

$P_4 = -VI$
$= -8 \times 0.2 \times 5$
$= -8$ W (supply)

$\therefore\ 48 + 60 - 100 - 8 = 0$
$\Rightarrow 108 = 108$

1.5 Given, (b)

$i = 5\cos 60\pi t$ A

$t = 5ms$

$= 5\times10^{-3}$ s

(a) $v = 2i$

(b) $v = \left(10 + 5\int_0^t i\,dt\right)$ v

(a) We know,

$P = VI$

$= 2\times5\cos 60\pi t \times 5\cos 60\pi t$

at $t = 5\times10^{-3}$ s,

$\therefore P = 2\times5\times\cos(60\times180\times5\times10^{-3})\times5\times\cos(60\times180\times5\times10^{-3})$

$= 17.27$ W [absorb power]

(b) We know,

$P = VI$

$= \left(10 + 5\int_0^t i\,dt\right)\times I$

$= \left[10 + 5\int_0^t 5\cos 60\pi t\,dt\right]\times 5\times\cos 60\pi t$

$= \left[10 + 5\times\frac{5\sin 60\pi t}{60\pi}\right]\times 5\cos 60\pi t$

at $t = 5\times10^{-3}$ s,

$\therefore P = \left[10 + 5\times\frac{5\times\sin(60\times180\times5\times10^{-3})}{60\times3.1416}\right]\times5\cos(60\times180\times5\times10^{-3})$

$= 29.7$ W [absorb power]

(Ans.)

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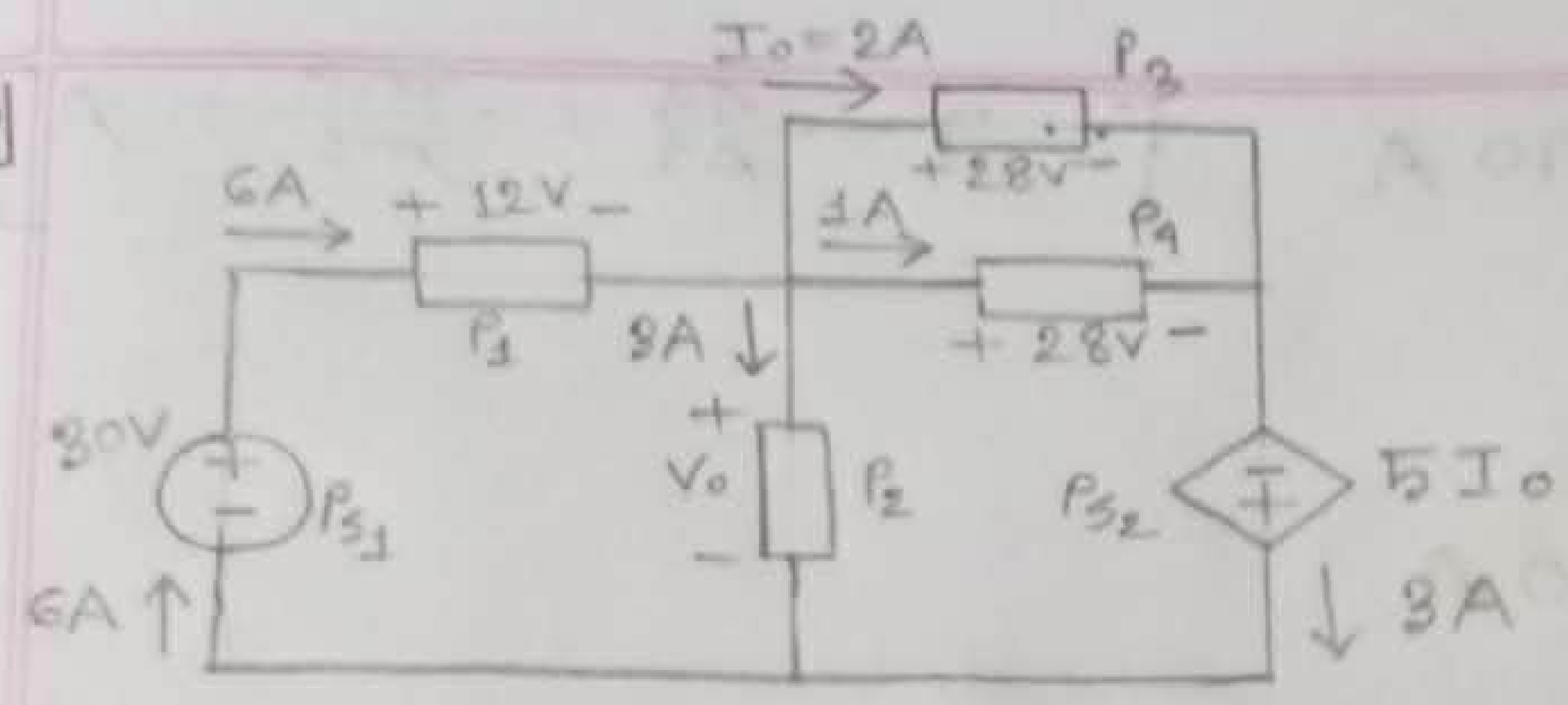
1.20



$P_{S_1} = -VI$
$= -30 \times 6$
$= -180$ W (supply)

$P_{S_2} = -VI$
$= -5 \times 2 \times 3$
$= -30$ W (supply)

$P_1 = +VI$
$= 12 \times 6$
$= 72$ W (absorb)

$P_2 = +VI$
$= V_o \times 3$
$= 3V_o$ (absorb)

$P_3 = +VI$
$= 28 \times 2$
$= 56$ W (absorb)

$P_4 = +VI$
$= 28 \times 1$
$= 28$ W (absorb)

$\therefore 72 + 3V_o + 56 + 28 - 180 - 30 = 0$

$\Rightarrow 156 + 3V_o = 210$

$\Rightarrow 3V_o = 210 - 156$

$\Rightarrow 3V_o = 54$

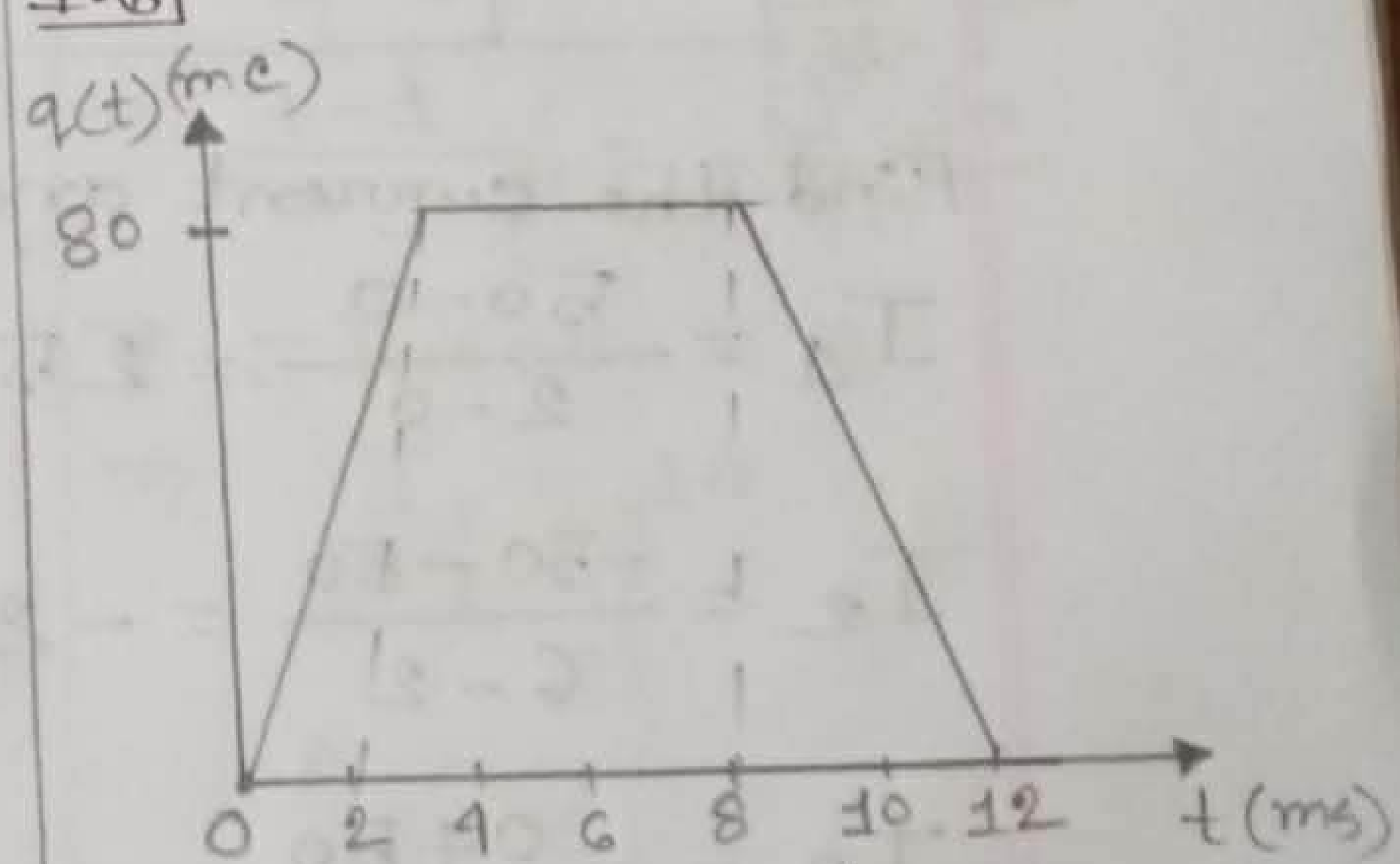
$\Rightarrow V_o = \frac{54}{3}$

$\Rightarrow V_o = 18$ V

(Ans.)

—— × ——

1.6



Find the current at:−

(a) $t = 1$ ms

$I = \frac{dq}{dt} = \frac{\Delta q}{\Delta t} = \frac{q_2 - q_1}{t_2 - t_1}$

$\left[\because I = \frac{\Delta q}{\Delta t} = \frac{mC}{ms} = A\right]$ 1.8

$$I = \frac{80-0}{2-0} = 40\ A$$

(b) at $t = 6\ ms$,

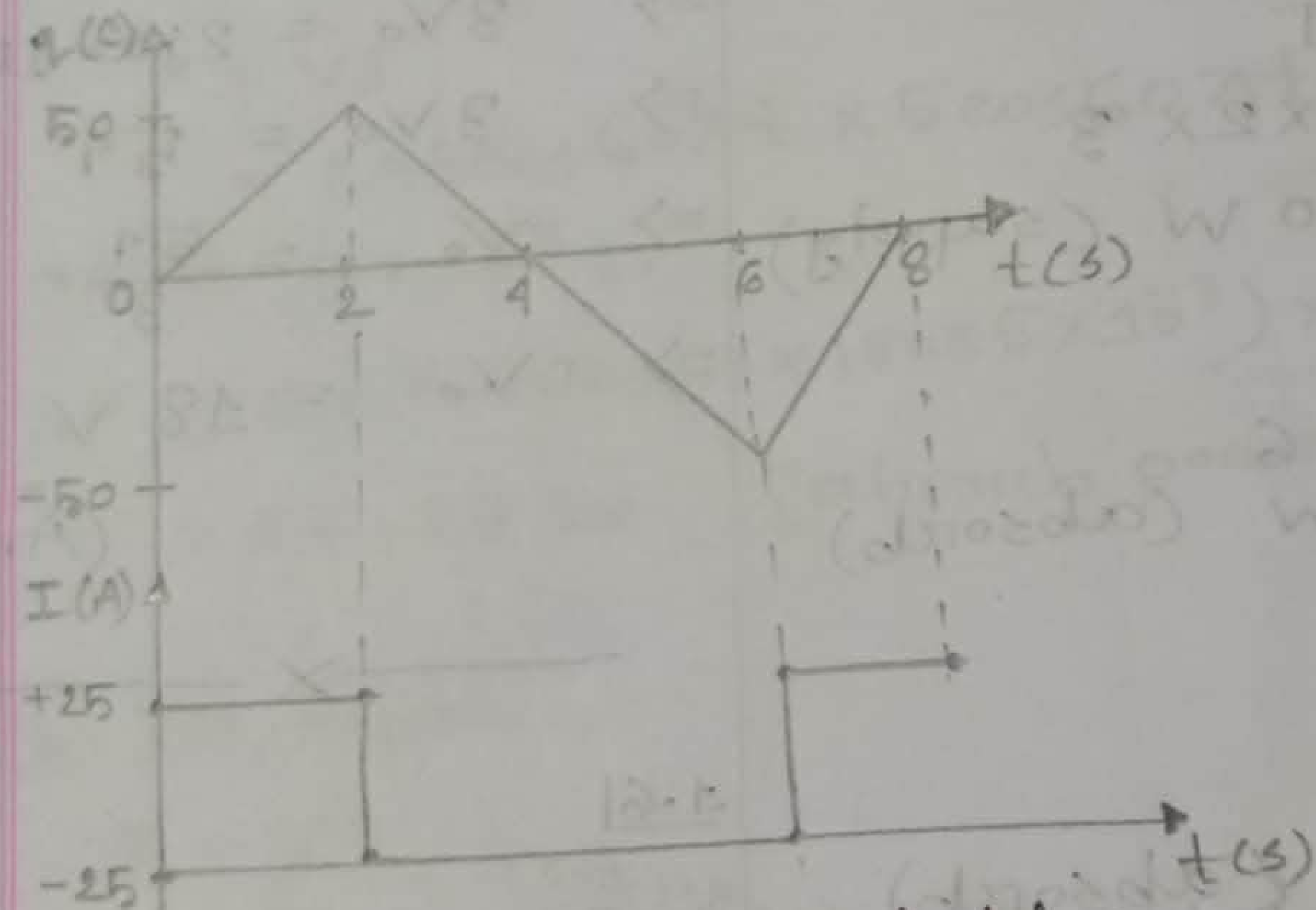
$$I = \frac{80-80}{8-2} = 0\ A$$

(c) at $t = 10\ ms$,

$$I = \frac{0-80}{12-8} = -20\ A$$

(Ans.)

1.7



Find the current and sketch :-

$$I_1 = \frac{50-0}{2-0} = 25\ A$$

$$I_2 = \frac{-50-50}{6-2} = -25\ A$$

$$I_3 = \frac{0+50}{8-6} = 25\ A$$

(Ans.)

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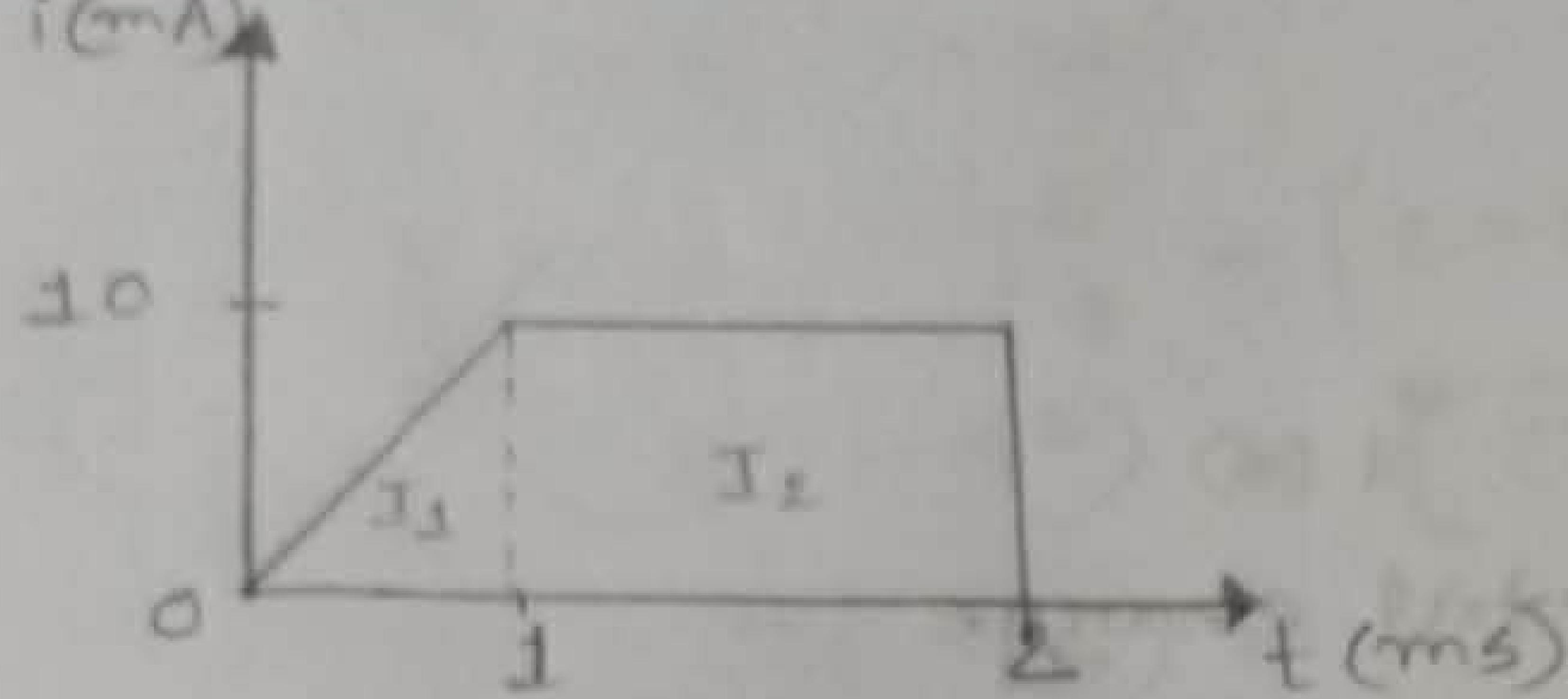
1.8 

Calculate the total charge:-

$$q = \int_{t_2}^{t_1} I\,dt$$

$$= \int_{t_1}^{t_2} I_1\,dt + \int_{t_2}^{t_3} I_2\,dt$$

$$= \int_0^1 10t\,dt + \int_1^2 10\,dt$$

$$= \left[10 \times \frac{t^2}{2}\right]_0^1 + \left[10t\right]_1^2$$

$$= (5 \times 1^2 - 5 \times 0^2) + (10 \times 2 - 10 \times 1)$$

$$= 5 + 10$$

$$= 15\ \mu C$$

$$\left[\because q = It = mA \times ms = 10^{-3} \times 10^{-3} = 10^{-6} \text{ (micro)}\right]$$

$$\frac{t - t_1}{t_1 - t_2} = \frac{I - I_1}{I_1 - I_2}$$

$I_1 \Rightarrow$

$$\Rightarrow \frac{t - 0}{0 - 1} = \frac{I - 0}{0 - 10}$$

$$\Rightarrow \frac{t}{-1} = \frac{I}{-10}$$

$$\Rightarrow -I = -10t$$

$$\Rightarrow I_1 = 10t$$

$I_2 \Rightarrow$

$$\frac{t - t_1}{t_1 - t_2} = \frac{I - I_1}{I_1 - I_2}$$

$$\Rightarrow \frac{t - 1}{1 - 2} = \frac{I - 10}{10 - 10}$$

$$\Rightarrow \frac{t - 1}{-1} = \frac{I - 10}{0}$$

$$\Rightarrow -I + 10 = 0$$

$$\Rightarrow I_2 = 10$$

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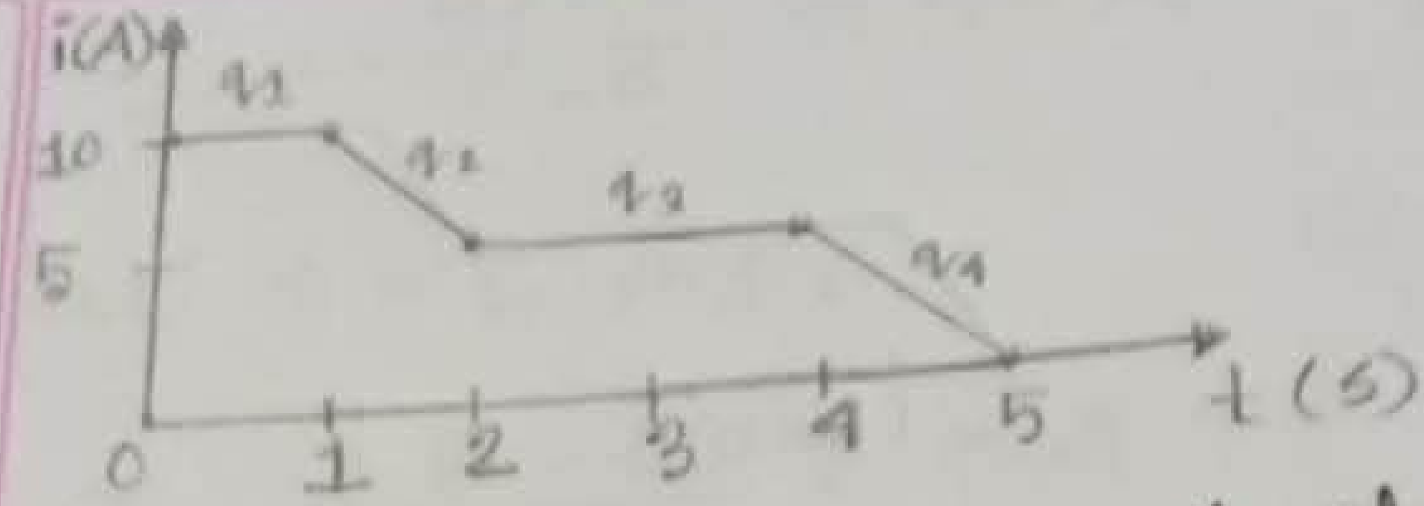
1.9



Determine the total charge :-

$$q_1 = \int_{t_1}^{t_2} I\,dt$$
$$= \int_0^1 10\,dt$$
$$= [10t]_0^1$$
$$= (10-0)$$
$$= 10\ C$$

$$q_2 = \int_{t_2}^{t_3} 5t\,dt$$
$$= \int_1^2 5t\,dt$$
$$= 5\left[\frac{t^2}{2}\right]_1^2$$
$$= \frac{5}{2}(4-1)$$
$$= \frac{15}{2}$$
$$= 7.5\ C$$

$$q_3 = \int_2^4 5\,dt$$
$$= [5t]_2^4$$
$$= (5\times4 - 5\times2)$$
$$= 20-10$$
$$= 10\ C$$

$$\frac{t-t_1}{t_1-t_2} = \frac{I-I_1}{I_1-I_2}$$

(0-1s)

$$\Rightarrow \frac{t-0}{0-1} = \frac{I-10}{10-10}$$
$$\Rightarrow \frac{t}{-1} = \frac{I-10}{0}$$
$$\Rightarrow -I+10 = 0$$
$$\Rightarrow I = 10\ A$$

(1-2s)

$$\Rightarrow \frac{t-1}{1-2} = \frac{I-5}{5-10}$$
$$\Rightarrow \frac{t-1}{-1} = \frac{I-5}{-5}$$
$$\Rightarrow -5t+5 = -I+5$$
$$\Rightarrow I = 5t\ A$$

(2-4s)

$$\Rightarrow \frac{t-2}{2-4} = \frac{I-5}{5-5}$$
$$\Rightarrow \frac{t-2}{-2} = \frac{I-5}{0}$$
$$\Rightarrow -2I+10 = 0$$
$$\Rightarrow I = 5\ A$$

(4-5s)

$$\Rightarrow \frac{t-4}{4-5} = \frac{I-0}{0-5}$$
$$\Rightarrow -5t+20 = -I$$
$$\Rightarrow I = 5t-20$$

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$q_4 = \int_4^5 (5t - 20)\,dt$

$= \left[5 \times \frac{t^2}{2}\right]_4^5 - \left[20t\right]_4^5$

$= \frac{5}{2}(5^2 - 4^2) - [20 \times 5 - 20 \times 4]$

$= \frac{5}{2}(25 - 16) - (100 - 80)$

$= 22.5 - 20$

$= 2.5$ C

(a) at $t = 1$ s,

$q = 10$ C

(b) at $t = 3$ s,

$q = 10 + 7.5 + 5$

$= 22.5$ C

(c) at $t = 5$ s,

$q = 10 + 7.5 + 10 + 2.5$

$= 30$ C

(Ans.)